Polar Form, De Moivre's Theorem & Roots of Unity

ANSWER KEY

Section 1: Polar and Euler Forms

1 Express $z = -2 - 2\sqrt{3}i$ in Euler's exponential form $re^{i\theta}$ with $\theta \in (-\pi, \pi]$.

a. $4e^{-i2\pi/3}$ ✓

b. $4e^{i4\pi/3}$

c. $2e^{-i2\pi/3}$

d. $4e^{-i\pi/3}$

$r = \sqrt{(-2)^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4$. In Quadrant III ($x < 0, y < 0$), $\theta = -\pi + \arctan(\sqrt{3}) = -\pi + \frac{\pi}{3} = -\frac{2\pi}{3}$. Thus $z = 4e^{-i2\pi/3}$.

2 Convert $z = 6e^{i3\pi/4}$ into exact Cartesian form $x + iy$.

a. $-3\sqrt{2} + 3\sqrt{2}i$ ✓

b. $3\sqrt{2} - 3\sqrt{2}i$

c. $-3 + 3\sqrt{3}i$

d. $-3\sqrt{2} - 3\sqrt{2}i$

$x = 6\cos(3\pi/4) = 6(-\sqrt{2}/2) = -3\sqrt{2}$. $y = 6\sin(3\pi/4) = 6(\sqrt{2}/2) = 3\sqrt{2}$.

3 If $z_1 = 3e^{i\pi/3}$ and $z_2 = 2e^{i\pi/6}$, evaluate the quotient $\frac{z_1}{z_2}$ in exponential form.

a. $\frac{3}{2}e^{i\pi/6}$ ✓

b. $\frac{3}{2}e^{i\pi/2}$

c. $6e^{i\pi/6}$

d. $1.5e^{-i\pi/6}$

$\frac{z_1}{z_2} = \frac{r_1}{r_2}e^{i(\theta_1 - \theta_2)} = \frac{3}{2}e^{i(\pi/3 - \pi/6)} = \frac{3}{2}e^{i\pi/6}$.

Section 2: De Moivre's Theorem for Powers

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4 Use De Moivre's Theorem to evaluate $(1 + i\sqrt{3})^6$.

a. 64 ✓

b. -64

c. $64i$

d. $-64i$

In polar form, $1 + i\sqrt{3} = 2e^{i\pi/3}$. By De Moivre's Theorem:
$(2e^{i\pi/3})^6 = 2^6 e^{i6(\pi/3)} = 64e^{i2\pi} = 64(\cos 2\pi + i\sin 2\pi) = 64(1 + 0) = 64$.

5 Evaluate $(\cos(\pi/5) + i\sin(\pi/5))^{10}$.

a. 1 ✓

b. -1

c. i

d. $-i$

By De Moivre's Theorem: $(\text{cis}(\pi/5))^{10} = \text{cis}(10 \times \pi/5) = \text{cis}(2\pi) = \cos(2\pi) + i\sin(2\pi) = 1$.

6 Express $\cos(3\theta)$ in terms of $\cos\theta$ using De Moivre's Theorem $(\cos\theta + i\sin\theta)^3$.

a. $4\cos^3\theta - 3\cos\theta$ ✓

b. $3\cos^3\theta - 4\cos\theta$

c. $\cos^3\theta - 3\cos\theta$

d. $4\cos^3\theta + 3\cos\theta$

Expand $(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$. Equating real parts:
$\cos(3\theta) = \cos^3\theta - 3\cos\theta(1 - \cos^2\theta) = 4\cos^3\theta - 3\cos\theta$.

Section 3: Roots of Complex Numbers

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7 Find all cube roots of $8i$ in polar form.

a. $2e^{i\pi/6}, 2e^{i5\pi/6}, 2e^{-i\pi/2}$ ✓

b. $2e^{i\pi/3}, 2e^{i\pi}, 2e^{i5\pi/3}$

c. $8e^{i\pi/6}, 8e^{i5\pi/6}, 8e^{-i\pi/2}$

d. $2,\ -1+i\sqrt{3},\ -1-i\sqrt{3}$

$8i = 8e^{i\pi/2}$. The cube roots have magnitude $8^{1/3} = 2$ and arguments $\frac{\pi/2 + 2k\pi}{3} = \frac{\pi}{6} + \frac{2k\pi}{3}$ for $k = 0, 1, 2$, giving $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{9\pi}{6} \equiv -\frac{\pi}{2}$.

8 How are the n distinct nth roots of any complex number $z = re^{i\theta}$ distributed on the Argand plane?

a. **They lie on a circle of radius $r^{1/n}$ centered at the origin, evenly spaced by $\frac{2\pi}{n}$ radians** ✓

b. They lie along a straight line through the origin

c. They form an ellipse with eccentricity $1/n$

d. They cluster exclusively in Quadrant I

All nth roots share the identical modulus $r^{1/n}$ and their arguments differ by constant increments of $\frac{2\pi}{n}$, forming a regular n-sided polygon centered at the origin.

9 What is the sum of all n distinct nth roots of any complex number z ($n \geq 2$)?

a. 0 ✓

b. 1

c. n

d. z

By Vieta's formulas applied to $w^n - z = 0$, the coefficient of w^{n-1} is zero, so the sum of all roots is strictly zero.

Section 4: Roots of Unity & Factorization

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10 If $\omega = e^{i2\pi/3}$ is a primitive complex cube root of unity, what is the value of $1 + \omega + \omega^2$?

a. 0 ✓

b. 1

c. -1

d. 3

For any nth root of unity $\omega \neq 1$, $\sum_{k=0}^{n-1} \omega^k = \frac{\omega^n - 1}{\omega - 1} = \frac{1-1}{\omega-1} = 0$. For $n = 3$, $1 + \omega + \omega^2 = 0$.

11 Factor $z^4 + 1$ into two real quadratic factors.

a. $(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$ ✓

b. $(z^2 + 1)^2$

c. $(z^2 - 1)(z^2 + 1)$

d. $(z^2 - 2z + 1)(z^2 + 2z + 1)$

Complete the square: $z^4 + 1 = (z^2 + 1)^2 - 2z^2 = (z^2 + 1)^2 - (\sqrt{2}z)^2 = (z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$.

12 The four 4th roots of unity ($z^4 = 1$) form the vertices of which geometric shape on the Argand plane?

a. **A square inscribed in the unit circle with vertices at $\pm 1, \pm i$** ✓

b. An equilateral triangle

c. A regular hexagon

d. A rectangle of perimeter 4

The solutions to $z^4 = 1$ are $1, i, -1, -i$. Plotted on the Argand diagram, they form a square of side length $\sqrt{2}$ inscribed in the unit circle.

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